

Algebra 1
Unit 5
Lesson 9 Practice

Name _____
Date _____
Period _____

Use the system of equations $\begin{cases} x + 4y = 3 \\ 0.5x - 4y = 6 \end{cases}$ to complete questions 1 – 4.

1. Complete the statement.

- ☐ Adding
☐ Subtracting

the original equations in this system would result in

eliminating one of the variables,

- ☐ x .
☐ y .

2. Perform the selected operation selected in question 1.

3. Solve the system using your equation from question 2.

4. Verify the solution algebraically using substitution.

5. Solve the system of equations using elimination.

$$\begin{cases} 5p + 2h = 13 \\ 3p - h = -12 \end{cases}$$



6. Verify the solution from question 5 algebraically using substitution.

7. Solve the system of equations algebraically.

$$\begin{cases} 3x + y = 5 \\ 5x - 4y = -12 \end{cases}$$

8. Verify the solution from question 7 algebraically using substitution.

9. Mia is solving the system of equations $\begin{cases} 2x + 3y = 7 \\ 3x - y = -6 \end{cases}$ using elimination. In order to eliminate the x , she multiplies the first equation by a factor of 3 and the second equation by a factor of 2. Explain whether her method will work.

10. Mia's system would be simpler to solve by eliminating the y . Describe why this is true and then solve Mia's system.